

object_state_sentence_focus

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load packages

```
library(tidyverse)
library(ggplot2)
library(ggthemes)
library(rstatix)
library(ggpubr)
library(emmeans)
library(dplyr)
library(lmerTest)
library(paletteer)
library(Hmisc)
```

import data

```
sona_data <- read.csv("../data/object-state-sentence-focus_SONA_clean_12.16.csv", fileEncoding = "UTF-8")
prolific_data <- read.csv("../data/object-state-sentence-focus-prolific_12.13.csv", fileEncoding = "UTF-8")

comb_data <- rbind(sona_data, prolific_data)
```

data treatment

```
## exclude non-English first language participants
lang <- comb_data %>% filter(english=="no")
comb_data <- comb_data %>% filter(!subject %in% lang$subject)

#how many participants
length(comb_data %>% pull(subject) %>% unique())
```

```
## [1] 230
```

comprehension scores

```
comprehension_scores <- comb_data %>% group_by(subject) %>%  
  filter(typeoftrial=="comprehension") %>% count(response==comprehension_type) %>%  
  pivot_wider(names_from = c(2), values_from = n) %>% rename(correct = "TRUE") %>%  
  rename(incorrect = "FALSE") %>% mutate(comp = correct /24)  
  
inaccurate_comp <- comprehension_scores %>% filter(comp < 0.5)  
  
#Dropping all p's with <50% on comprehension score  
comb_data<- comb_data %>% filter(!(subject %in% inaccurate_comp$subject))  
  
#How many with good comprehension  
length(comb_data %>% pull(subject) %>% unique())
```

```
## [1] 230
```

exclusions

```
##exclude filler, practice, and non-picture trials  
exclusions_data <- comb_data %>%  
  filter(typeoftrial=="picture" & !(trialcondition %in% c("filler", "practice_y","practice_n"))) %>%  
  ##calculate subject accuracy  
  mutate(accuracy=ifelse(correct=="TRUE",1,0)) %>%  
  separate(trialcondition, into = c("weight", "state","focus"))
```

item labels

```
items <- gsub('.{0,5}$', '', exclusions_data$object) %>% as.data.frame()  
exclusions_data <- exclusions_data %>% mutate(item = items[,1] )  
  
### For experimental picture trials: this code creates a new column 'items' to id the item (picture sti
```

accuracy analysis

```
##accuracy check
```

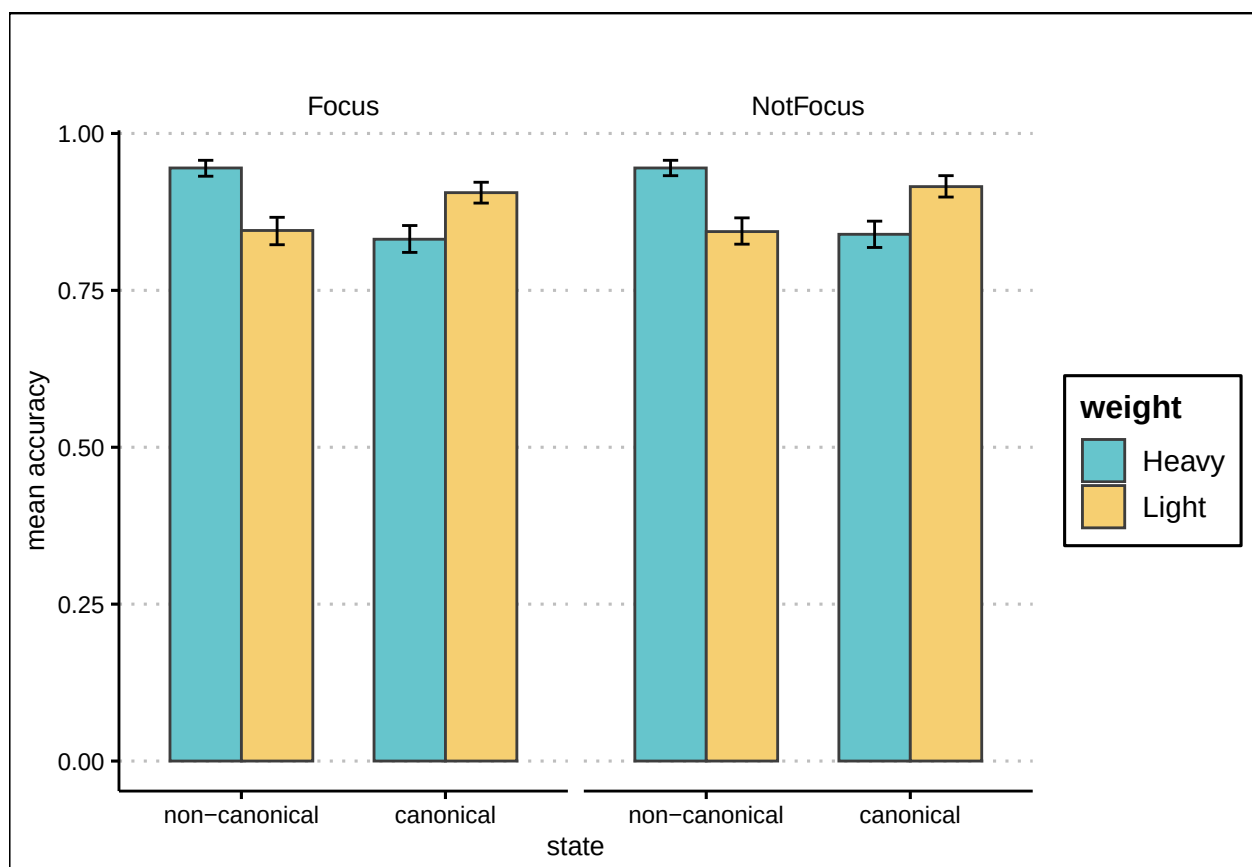
```
#marking subjects w <80 % accuracy on target trials  
subject_accuracy <- exclusions_data %>% group_by(subject) %>% summarise(TaskAccuracy=mean(accuracy))  
InaccurateSubjects <- subject_accuracy %>% filter(TaskAccuracy < 0.8)
```

plot

```

exclusions_data %>%
  group_by(weight, state, focus) %>%
  summarise(ci = list(mean_cl_boot(accuracy) %>%
    rename(mean=y, lwr=ymin, upr=ymax))) %>% unnest%>%
  mutate(state = fct_recode(state, canonical = "Normal", `non-canonical` = "Smashed" ),
    state = fct_relevel(state, "non-canonical", "canonical"))%>%
  ggplot(aes(x= state, y = mean, group = weight, fill = weight)) + facet_wrap(~focus)+
  geom_bar(stat = "identity", position = "dodge", width = 0.7, color = "gray24")+
  geom_errorbar(aes(ymin=lwr, ymax=upr), size = 0.5, width=.15,
    color = "black", position = position_dodge(0.70))+
  labs(y = "mean accuracy", title = "")+
  theme_clean()+
  scale_fill_paletteer_d("rcartocolor::Pastel")

```



main accuracy model

```

#maximum converging model
acc_model = glmer(data = exclusions_data,
  accuracy ~ state*weight*focus +
    (1 + state+weight+focus|subject) + (1|stimulus),
  family = "binomial",
  control = glmerControl(optimizer = "bobyqa",
    optCtrl=list(maxfun=200000),
    check.conv.singular = .makeCC(action = "ignore", tol = 1e-4)))
summary(acc_model)

```

```

## Generalized linear mixed model fit by maximum likelihood (Laplace
## Approximation) [glmerMod]
## Family: binomial ( logit )
## Formula: accuracy ~ state * weight * focus + (1 + state + weight + focus |
## subject) + (1 | stimulus)
## Data: exclusions_data
## Control: glmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
## check.conv.singular = .makeCC(action = "ignore", tol = 1e-04))
##
##      AIC      BIC    logLik deviance df.resid
##  4293.9   4429.3  -2128.0   4255.9     9141
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -18.1642   0.0493   0.1120   0.2376   7.0996
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## subject (Intercept) 6.9514 2.6365
##          stateSmashed 1.5240 1.2345 -0.17
##          weightLight 1.2315 1.1097 -0.21 -0.91
##          focusNotFocus 0.2530 0.5030 -0.13 -0.76 0.68
## stimulus (Intercept) 0.1827 0.4275
## Number of obs: 9160, groups: subject, 229; stimulus, 80
##
## Fixed effects:
##
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)      3.03765    0.25956  11.703 < 2e-16 ***
## stateSmashed      2.21899    0.31807   6.976 3.03e-12 ***
## weightLight       1.23347    0.27148   4.543 5.53e-06 ***
## focusNotFocus      0.08047    0.19847   0.405 0.685
## stateSmashed:weightLight -3.60090    0.32290 -11.152 < 2e-16 ***
## stateSmashed:focusNotFocus -0.30823    0.29605  -1.041 0.298
## weightLight:focusNotFocus 0.24651    0.25876   0.953 0.341
## stateSmashed:weightLight:focusNotFocus -0.09854    0.38211  -0.258 0.796
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) sttSms wghtLg fcsNtF sttS:L stS:NF wgL:NF
## stateSmashd -0.217
## weightLight -0.266 -0.102
## focusNotFcs -0.377 0.072 0.266
## sttSmsd:wL 0.030 -0.578 -0.503 -0.121
## sttSmsd:NF 0.116 -0.466 -0.154 -0.329 0.456
## wghtLght:NF 0.160 -0.108 -0.471 -0.412 0.358 0.282
## sttSms:L:NF -0.076 0.327 0.317 0.202 -0.584 -0.741 -0.648

```

```
car::Anova(acc_model)
```

```

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: accuracy
##
##              Chisq Df Pr(>Chisq)

```

```
## state          0.0023  1    0.96184
## weight         0.6177  1    0.43190
## focus          0.0335  1    0.85469
## state:weight   193.8344  1    < 2e-16 ***
## state:focus    3.3711  1    0.06635 .
## weight:focus   1.0631  1    0.30252
## state:weight:focus 0.0665  1    0.79650
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(acc_model, pairwise~weight*state, simple="weight")
```

```
## $emmeans
## weight state emmean SE df asymp.LCL asymp.UCL
## Heavy Normal 3.08 0.240 Inf 2.61 3.55
## Light Normal 4.43 0.303 Inf 3.84 5.03
## Heavy Smashed 5.14 0.324 Inf 4.51 5.78
## Light Smashed 2.85 0.207 Inf 2.44 3.25
##
## Results are averaged over the levels of: focus
## Results are given on the logit (not the response) scale.
## Confidence level used: 0.95
##
## $contrasts
## state = Normal:
## contrast estimate SE df z.ratio p.value
## Heavy - Light -1.36 0.240 Inf -5.664 <.0001
##
## state = Smashed:
## contrast estimate SE df z.ratio p.value
## Heavy - Light 2.29 0.262 Inf 8.750 <.0001
##
## Results are averaged over the levels of: focus
## Results are given on the log odds ratio (not the response) scale.
```

RT analyses

exclude inaccurate subjects

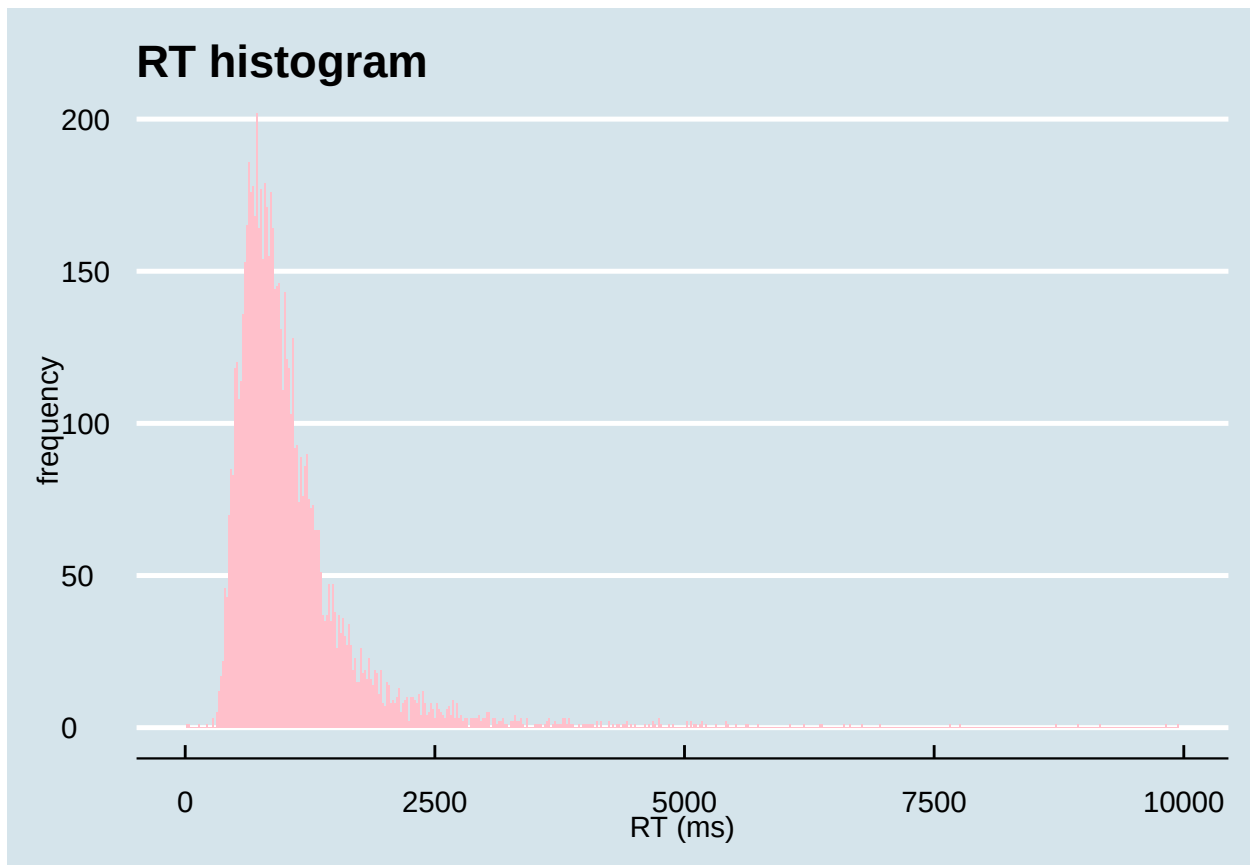
```
##exclude incorrect responses
rt_data <- exclusions_data %>% filter(accuracy==1) %>%
  ##exclude subjects with accuracy < 80%
  filter(!subject %in% InaccurateSubjects$subject)

##total participants after exclusion: N = 186
length(rt_data %>% pull(subject) %>% unique())
```

```
## [1] 186
```

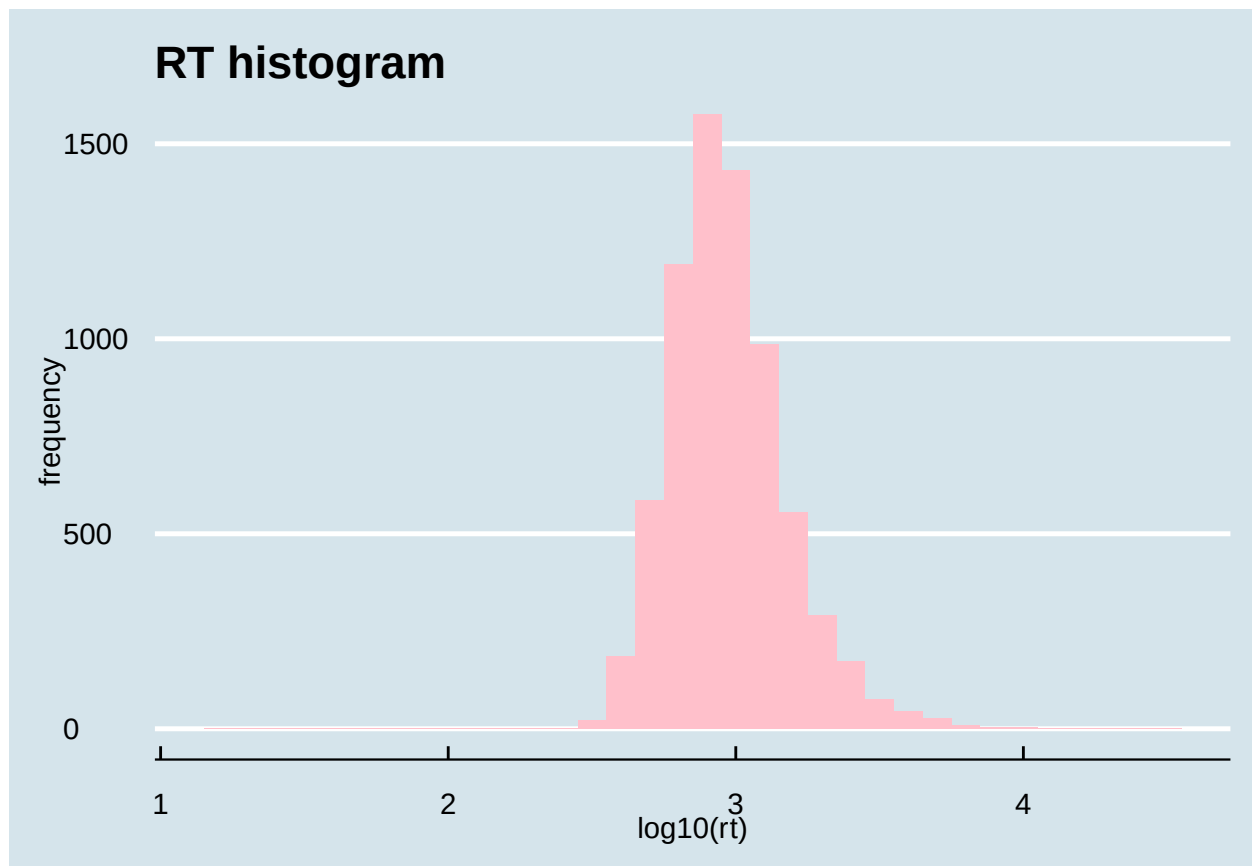
transforming RT

```
##make rt numeric
rt_data <- rt_data %>% mutate(rt = as.numeric(rt))
## rt histogram ----does NOT look normally distributed (positive skew)
rt_data %>% filter(rt <10000) %>%
  ggplot(aes(x=rt)) + geom_histogram(binwidth =20, fill="pink" ) +
  theme_economist() + labs(x="RT (ms)", y="frequency", title= "RT histogram")
```



```
##rt log (base 10) transformation
rt_data <- rt_data %>% mutate(log10_rt=log(rt, 10))

##log10(rt) histogram
rt_data %>%
  ggplot(aes(x=log10_rt)) + geom_histogram(binwidth =0.1, fill="pink" ) +
  theme_economist() + labs(x="log10(rt)", y="frequency", title= "RT histogram")
```



MAD exclusions

```
mad_data = rt_data %>%
  group_by(weight, state, focus) %>%
  mutate(MAD=3*mad(log10_rt), med=median(log10_rt),
         L= (med - MAD), U=(med + MAD)) %>%
  filter(log10_rt < U) %>% filter(log10_rt > L)

#percent data excluded
(nrow(rt_data)-nrow(mad_data))/nrow(rt_data)
```

```
## [1] 0.01700348
```

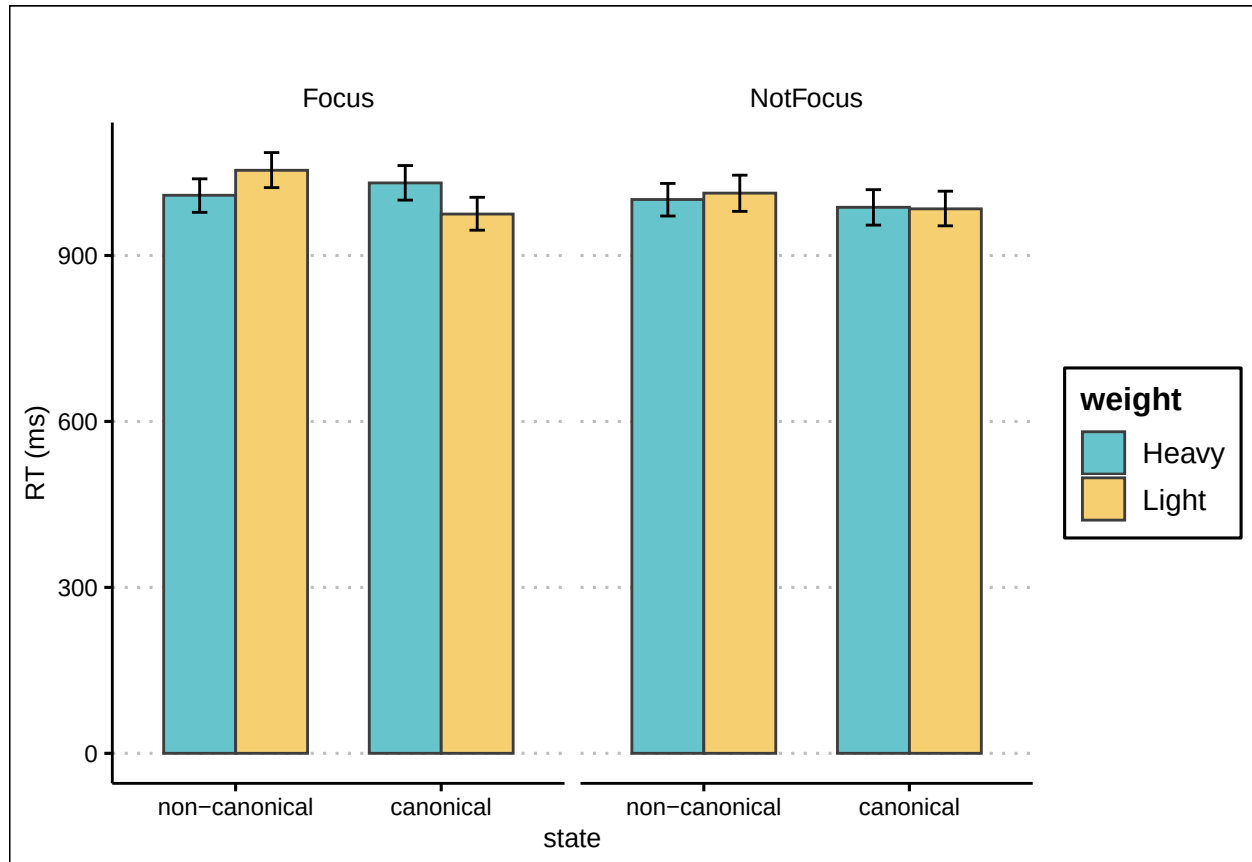
plot

```
mad_data %>%
  group_by(weight, state, focus) %>%
  summarise(ci = list(mean_cl_boot(rt) %>%
                      rename(mean=y, lwr=ymin, upr=ymax))) %>% unnest%>%
  mutate(state = fct_recode(state, canonical = "Normal", `non-canonical` = "Smashed" ),
         state = fct_relevel(state, "non-canonical", "canonical"))%>%
```

```

ggplot(aes(x= state, y = mean, group = weight, fill = weight)) + facet_wrap(~focus)+
geom_bar(stat = "identity", position = "dodge", width = 0.7, color = "gray24")+
geom_errorbar(aes(ymin=lwr, ymax=upr), size = 0.5, width=.15,
              color = "black", position = position_dodge(0.70))+
labs(y = "RT (ms)", title = "")+
theme_clean()+
scale_fill_paletteer_d("rcartocolor::Pastel")

```



main RT model

```

#maximum converging model
#failed to converge when including slope for focus
rt_model = lmer(data = mad_data,
                log10_rt ~ weight*state*focus +
                  (1 + weight+state|subject) + (1|stimulus),
                REML = FALSE,
                lmerControl(optimizer = "bobyqa",
                             optCtrl=list(maxfun=200000),
                             check.conv.singular = .makeCC(action = "ignore", tol = 1e-4)))
summary(rt_model)

```

```

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]

```

```

## Formula: log10_rt ~ weight * state * focus + (1 + weight + state | subject) +
## (1 | stimulus)
## Data: mad_data
## Control: lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
## check.conv.singular = .makeCC(action = "ignore", tol = 1e-04))
##
##      AIC      BIC    logLik deviance df.resid
## -7800.3 -7690.5  3916.2 -7832.3    7037
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.7635 -0.6266 -0.1018  0.5171  5.5469
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## subject (Intercept) 1.428e-02 0.119502
##          weightLight 1.652e-05 0.004064 0.05
##          stateSmashed 3.698e-04 0.019231 0.06 1.00
## stimulus (Intercept) 2.369e-03 0.048676
## Residual 1.700e-02 0.130381
## Number of obs: 7053, groups: subject, 186; stimulus, 80
##
## Fixed effects:
##
##              Estimate Std. Error      df t value
## (Intercept)      2.975e+00  1.247e-02  2.643e+02  238.569
## weightLight      -2.098e-02  6.265e-03  5.827e+03  -3.349
## stateSmashed      -5.518e-03  1.261e-02  1.202e+02  -0.437
## focusNotFocus     -1.619e-02  6.247e-03  6.635e+03  -2.591
## weightLight:stateSmashed  3.701e-02  8.833e-03  6.627e+03   4.190
## weightLight:focusNotFocus  1.893e-02  8.845e-03  6.632e+03   2.140
## stateSmashed:focusNotFocus  1.341e-02  8.803e-03  6.625e+03   1.524
## weightLight:stateSmashed:focusNotFocus -3.481e-02  1.250e-02  6.628e+03  -2.785
##
##              Pr(>|t|)
## (Intercept)      < 2e-16 ***
## weightLight      0.000815 ***
## stateSmashed      0.662577
## focusNotFocus      0.009588 **
## weightLight:stateSmashed  2.83e-05 ***
## weightLight:focusNotFocus  0.032365 *
## stateSmashed:focusNotFocus  0.127675
## weightLight:stateSmashed:focusNotFocus 0.005367 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##      (Intr) wghtLg sttSms fcsNtF wghL:S wgL:NF stS:NF
## weightLight -0.248
## stateSmashed -0.495 0.252
## focusNotFcs -0.251 0.501 0.248
## wghtLght:sS  0.177 -0.708 -0.348 -0.355
## wghtLght:NF  0.178 -0.710 -0.176 -0.709 0.504
## sttSmsd:NF  0.178 -0.356 -0.349 -0.710 0.500 0.503
## wghtLg:S:NF -0.126 0.503 0.247 0.501 -0.710 -0.708 -0.706

```

```
car::Anova(rt_model)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: log10_rt
##               Chisq Df Pr(>Chisq)
## weight         0.2890  1  0.590882
## state           0.9151  1  0.338764
## focus           7.7475  1  0.005379 **
## weight:state     9.8708  1  0.001679 **
## weight:focus     0.0574  1  0.810660
## state:focus      0.3928  1  0.530808
## weight:state:focus 7.7566  1  0.005352 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(rt_model, pairwise~weight*state*focus, simple="weight")
```

```
## $emmeans
##   weight state   focus   emmean    SE df asymp.LCL asymp.UCL
## Heavy  Normal Focus      2.98 0.0125 Inf      2.95      3.00
## Light  Normal Focus      2.95 0.0125 Inf      2.93      2.98
## Heavy  Smashed Focus      2.97 0.0126 Inf      2.94      2.99
## Light  Smashed Focus      2.99 0.0127 Inf      2.96      3.01
## Heavy  Normal NotFocus    2.96 0.0125 Inf      2.93      2.98
## Light  Normal NotFocus    2.96 0.0125 Inf      2.93      2.98
## Heavy  Smashed NotFocus    2.97 0.0126 Inf      2.94      2.99
## Light  Smashed NotFocus    2.97 0.0127 Inf      2.94      2.99
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## state = Normal, focus = Focus:
##   contrast      estimate      SE df z.ratio p.value
## Heavy - Light  0.020983 0.00626 Inf   3.349  0.0008
##
## state = Smashed, focus = Focus:
##   contrast      estimate      SE df z.ratio p.value
## Heavy - Light -0.016027 0.00624 Inf  -2.568  0.0102
##
## state = Normal, focus = NotFocus:
##   contrast      estimate      SE df z.ratio p.value
## Heavy - Light  0.002052 0.00623 Inf   0.330  0.7417
##
## state = Smashed, focus = NotFocus:
##   contrast      estimate      SE df z.ratio p.value
## Heavy - Light -0.000149 0.00624 Inf  -0.024  0.9810
##
## Degrees-of-freedom method: asymptotic
```

power analyses

```
#library(simr)
#set.seed(123)
# Change the fixed effect to be small
#fixef(rt_model)["weightLight:stateSmashed:focusNotFocus"] = 0.05
# Start with an extended model up to 250 participants
#extended_model <- extend(rt_model, along = "subject", n = 250)
# Perform a power curve analysis to test the three-way interaction
#power_curve <- powerCurve(extended_model,
#                           along = "subject",
#                           fixed("weight:state:focus", "anova"),
#                           nsim = 100) # Number of simulations per sample size

# Plot the power curve
#plot(power_curve)

#commenting out since it takes a while
```

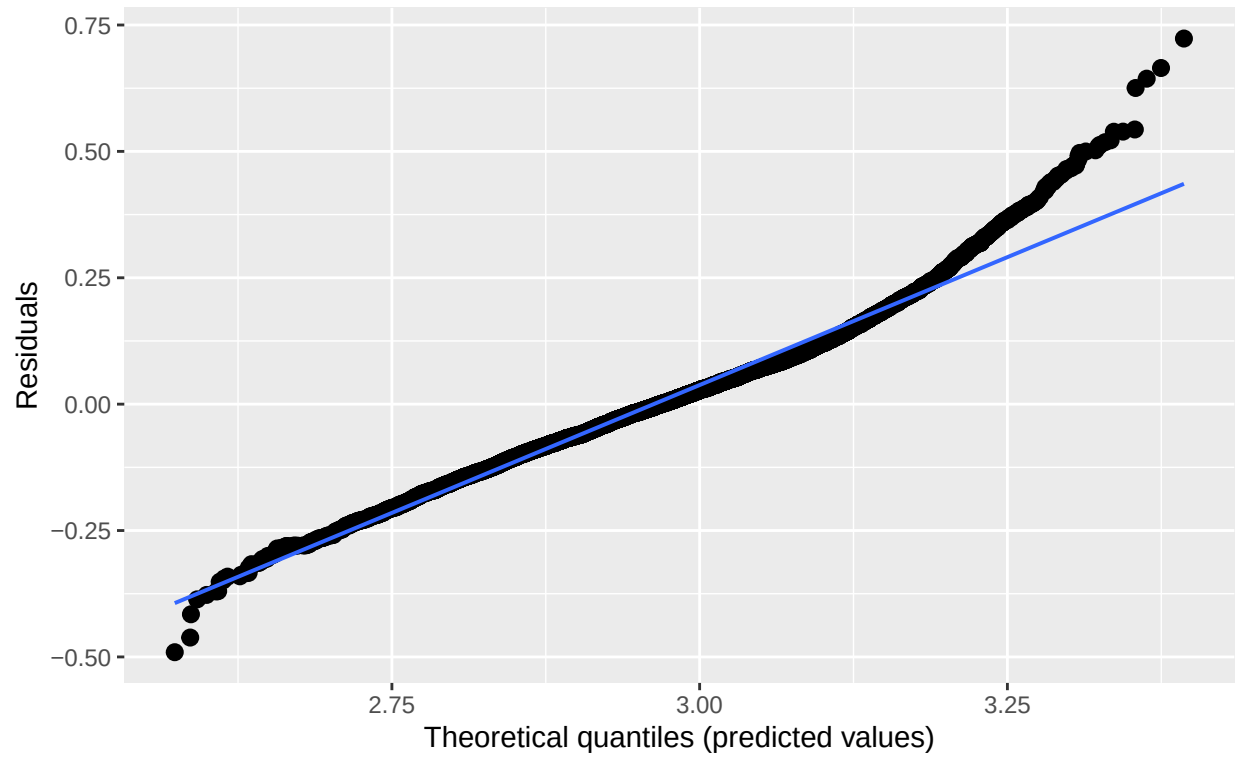
assumptions

```
library(glmTMB)
sjPlot::plot_model(rt_model, type='diag')
```

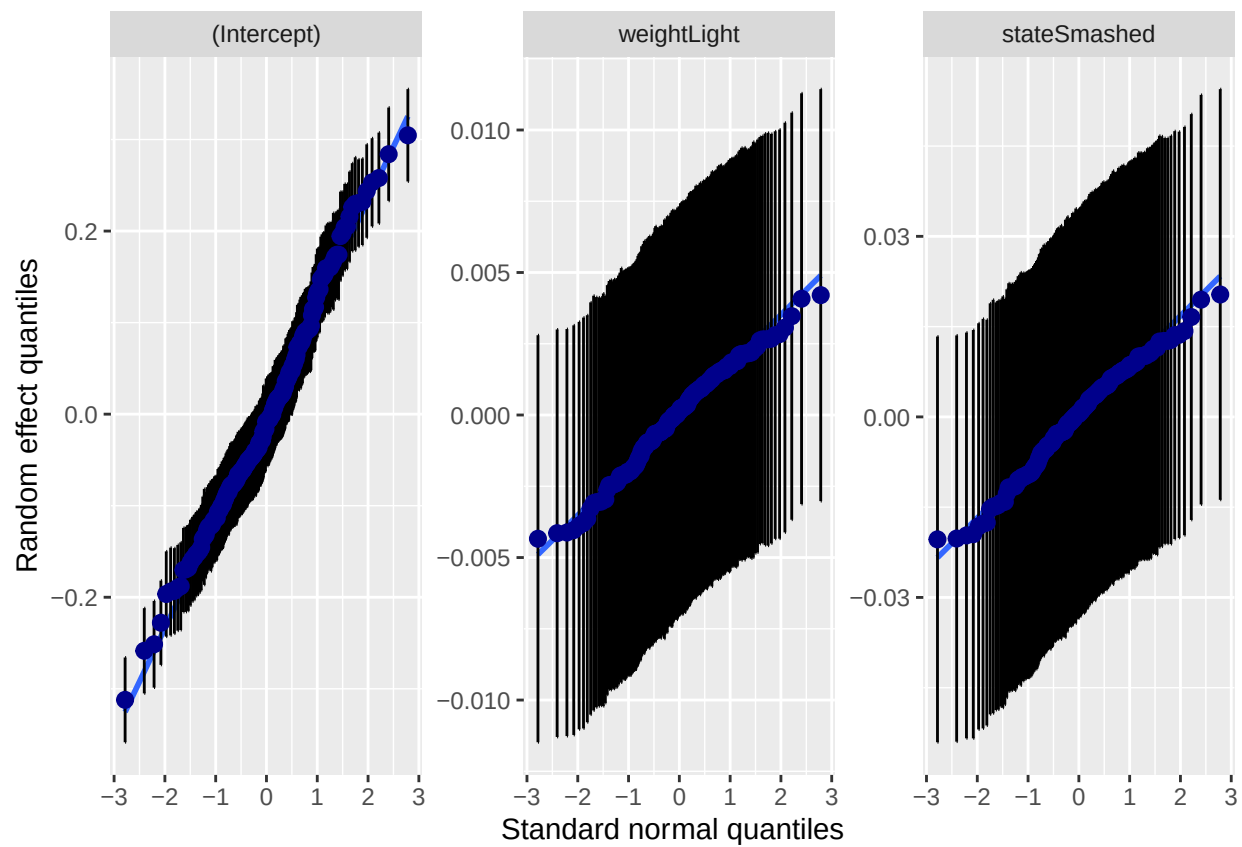
```
## [[1]]
```

Non-normality of residuals and outliers

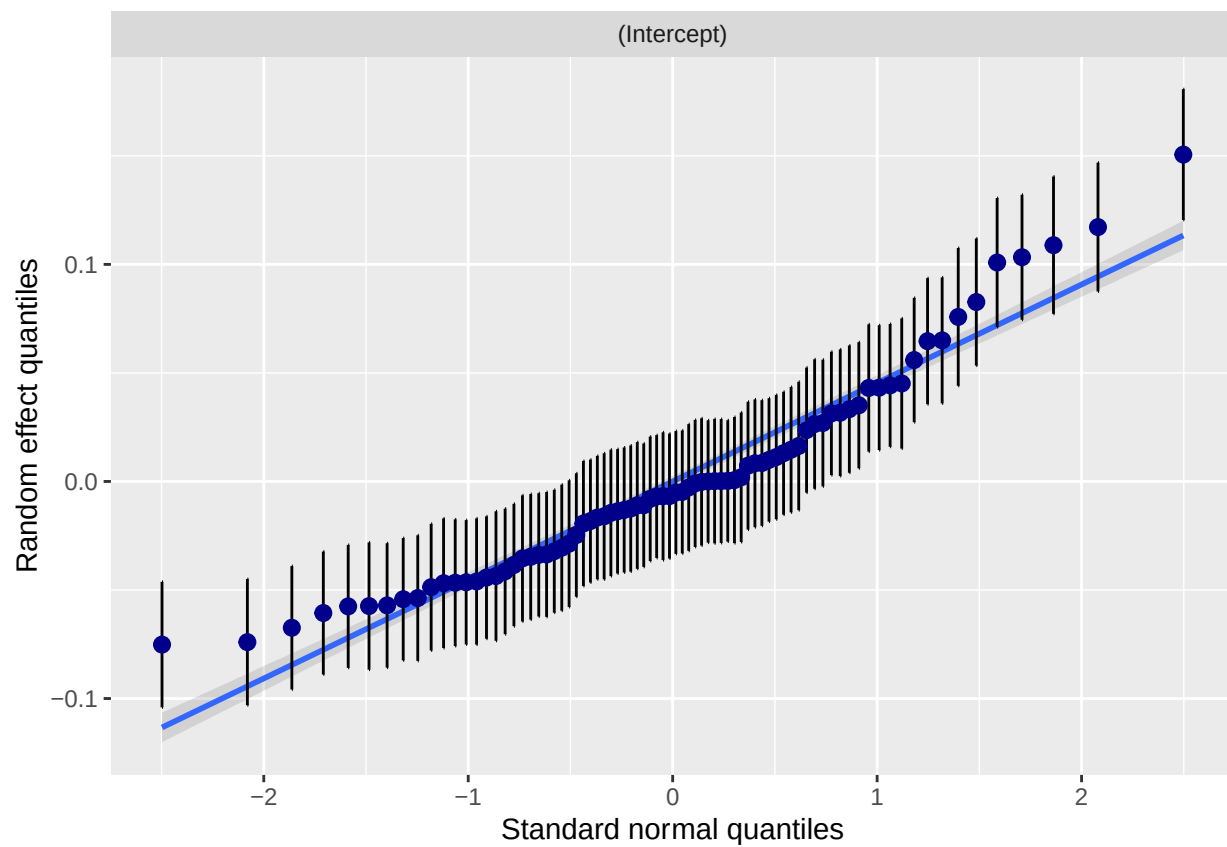
Dots should be plotted along the line



```
##  
## [[2]]  
## [[2]]$subject
```



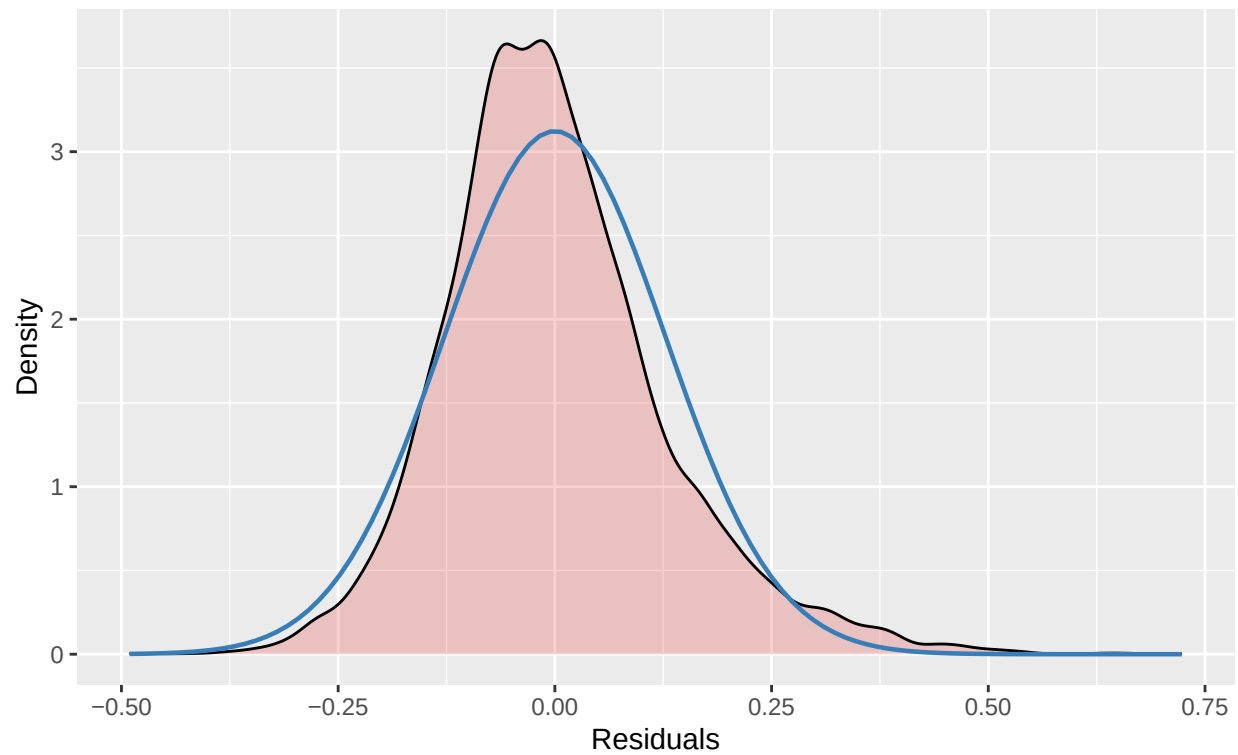
```
##
## [[2]]$stimulus
```



```
##
##
## [[3]]
```

Non-normality of residuals

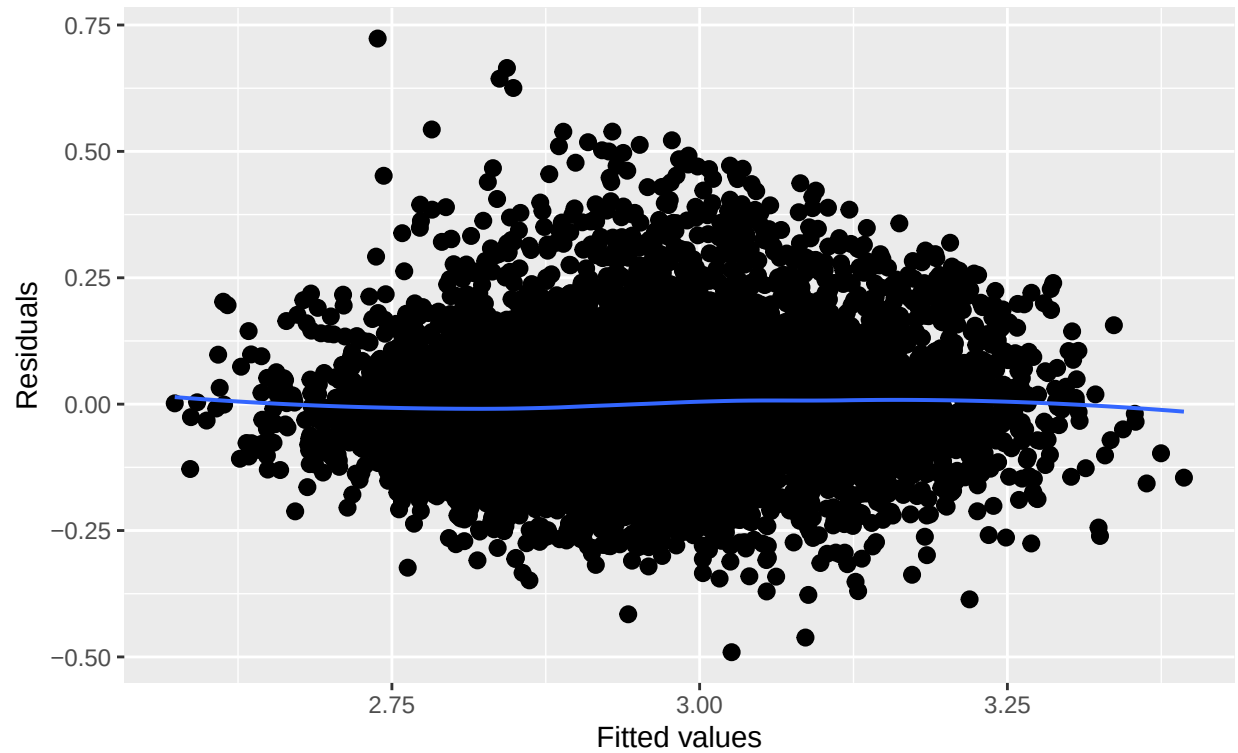
Distribution should look like normal curve



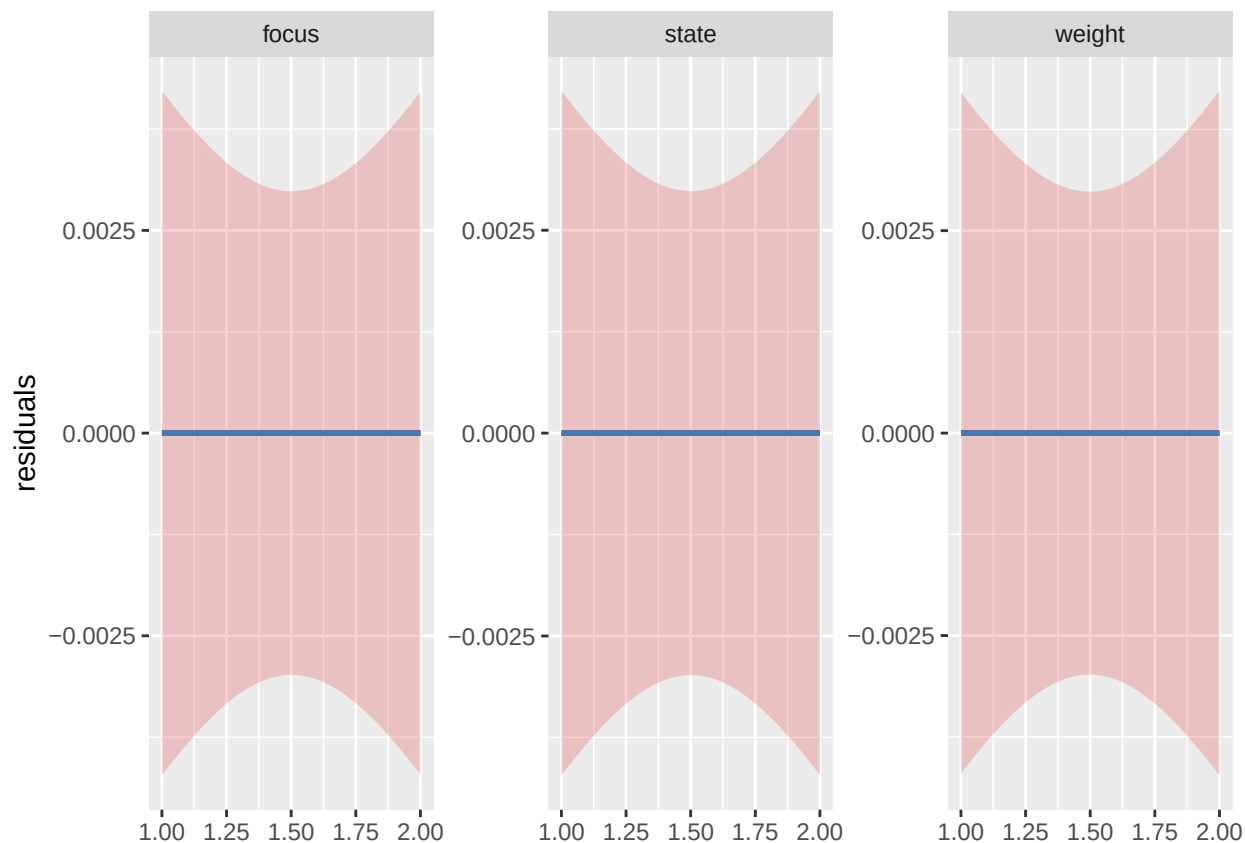
```
##  
## [[4]]
```

Homoscedasticity (constant variance of residuals)

Amount and distance of points scattered above/below line is equal or randomly spread



```
sjPlot::plot_model(rt_model, type='resid')
```



posthoc models

```
# split into picture type (state)
intact_data = mad_data %>% filter(state == "Normal")
squashed_data = mad_data %>% filter(state == "Smashed")
#run separate models on these data
intact_model = lmer(data = intact_data, log10_rt ~ weight*focus + (weight+focus | subject) +
  (1 | item), REML = FALSE,
  lmerControl(optimizer = "bobyqa",
    optCtrl=list(maxfun=200000),
    check.conv.singular = .makeCC(action = "ignore", tol = 1e-4)))

summary(intact_model)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log10_rt ~ weight * focus + (weight + focus | subject) + (1 |
## item)
## Data: intact_data
## Control: lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
## check.conv.singular = .makeCC(action = "ignore", tol = 1e-04))
##
##      AIC      BIC    logLik deviance df.resid
## -3671.4 -3597.4   1847.7  -3695.4     3511
```

```
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.5476 -0.6130 -0.0906  0.5116  5.4653
##
## Random effects:
##   Groups   Name                Variance Std.Dev. Corr
##   subject  (Intercept)         1.352e-02 0.116288
##             weightLight        4.142e-05 0.006436 0.02
##             focusNotFocus       2.020e-04 0.014213 0.41 0.92
##   item     (Intercept)         2.913e-03 0.053971
##   Residual                    1.710e-02 0.130752
## Number of obs: 3523, groups:  subject, 186; item, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept)      2.975e+00  1.285e-02  1.384e+02  231.456 < 2e-16 ***
## weightLight      -2.067e-02  6.299e-03  1.897e+03  -3.282  0.00105 **
## focusNotFocus    -1.600e-02  6.355e-03  8.057e+02  -2.518  0.01199 *
## weightLight:focusNotFocus  1.864e-02  8.877e-03  3.135e+03   2.100  0.03580 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##              (Intr) wghtLg fcsNtF
## weightLight  -0.241
## focusNotFcs  -0.196  0.505
## wghtLght:NF   0.173 -0.709 -0.699
```

```
car::Anova(intact_model)
```

```
## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: log10_rt
##              Chisq Df Pr(>Chisq)
## weight        6.4654  1    0.01100 *
## focus         2.1565  1    0.14197
## weight:focus  4.4102  1    0.03572 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(intact_model, pairwise~weight*focus, simple="weight")
```

```
## $emmeans
##   weight focus   emmean      SE df asymp.LCL asymp.UCL
## Heavy  Focus    2.97 0.0129 Inf      2.95      3.00
## Light  Focus    2.95 0.0129 Inf      2.93      2.98
## Heavy  NotFocus  2.96 0.0132 Inf      2.93      2.98
## Light  NotFocus  2.96 0.0132 Inf      2.93      2.98
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
```

```

## $contrasts
## focus = Focus:
## contrast      estimate      SE df z.ratio p.value
## Heavy - Light  0.02067 0.00630 Inf   3.282  0.0010
##
## focus = NotFocus:
## contrast      estimate      SE df z.ratio p.value
## Heavy - Light  0.00203 0.00626 Inf   0.325  0.7455
##
## Degrees-of-freedom method: asymptotic

squashed_model = lmer(data = squashed_data, log10_rt ~ weight*focus + (weight+focus | subject) +
                      (1 | item), REML = FALSE,
                      lmerControl(optimizer = "bobyqa",
                                   optCtrl=list(maxfun=200000),
                                   check.conv.singular = .makeCC(action = "ignore", tol = 1e-4)))
summary(squashed_model)

## Linear mixed model fit by maximum likelihood . t-tests use Satterthwaite's
## method [lmerModLmerTest]
## Formula: log10_rt ~ weight * focus + (weight + focus | subject) + (1 |
## item)
## Data: squashed_data
## Control: lmerControl(optimizer = "bobyqa", optCtrl = list(maxfun = 2e+05),
## check.conv.singular = .makeCC(action = "ignore", tol = 1e-04))
##
##      AIC      BIC    logLik deviance df.resid
## -3747.3 -3673.3  1885.7  -3771.3     3518
##
## Scaled residuals:
##      Min       1Q   Median       3Q      Max
## -3.0421 -0.6224 -0.1104  0.5119  4.2204
##
## Random effects:
## Groups Name Variance Std.Dev. Corr
## subject (Intercept) 1.549e-02 0.124468
##          weightLight 3.089e-05 0.005558 -0.18
##          focusNotFocus 1.066e-04 0.010322 -0.26 1.00
## item (Intercept) 1.742e-03 0.041731
## Residual 1.681e-02 0.129666
## Number of obs: 3530, groups: subject, 186; item, 40
##
## Fixed effects:
##              Estimate Std. Error      df t value Pr(>|t|)
## (Intercept) 2.970e+00 1.208e-02 1.862e+02 245.862 <2e-16 ***
## weightLight 1.581e-02 6.215e-03 1.969e+03 2.544 0.0110 *
## focusNotFocus -2.995e-03 6.216e-03 8.567e+02 -0.482 0.6301
## weightLight:focusNotFocus -1.563e-02 8.784e-03 3.149e+03 -1.779 0.0753 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Correlation of Fixed Effects:
##          (Intr) wghtLg fcsNtF
## weightLight -0.262

```

```
## focusNotFcs -0.278 0.502
## wghtLght:NF 0.180 -0.707 -0.699

car::Anova(squashed_model)

## Analysis of Deviance Table (Type II Wald chisquare tests)
##
## Response: log10_rt
##           Chisq Df Pr(>Chisq)
## weight      3.3094 1    0.06888 .
## focus        5.8122 1    0.01592 *
## weight:focus 3.1650 1    0.07523 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
emmeans(squashed_model, pairwise~weight*focus, simple="weight")
```

```
## $emmeans
##   weight focus   emmean      SE df asymp.LCL asymp.UCL
## Heavy  Focus    2.97 0.0121 Inf      2.95      2.99
## Light  Focus    2.99 0.0120 Inf      2.96      3.01
## Heavy  NotFocus  2.97 0.0120 Inf      2.94      2.99
## Light  NotFocus  2.97 0.0119 Inf      2.94      2.99
##
## Degrees-of-freedom method: asymptotic
## Confidence level used: 0.95
##
## $contrasts
## focus = Focus:
## contrast      estimate      SE df z.ratio p.value
## Heavy - Light -0.015814 0.00622 Inf  -2.544 0.0109
##
## focus = NotFocus:
## contrast      estimate      SE df z.ratio p.value
## Heavy - Light -0.000187 0.00621 Inf  -0.030 0.9760
##
## Degrees-of-freedom method: asymptotic
```

Demographics and other Descriptives

Experiment Duration (minutes)

```
comb_data<- comb_data %>% filter(!(subject %in% InaccurateSubjects$subject))
duration <- comb_data %>% filter(typeoftrial == "english") %>%
  mutate(time_elapsed = time_elapsed/1000/60)

duration %>% summarise(med = median(time_elapsed), iqr = IQR(time_elapsed), max = max(time_elapsed), min = min(time_elapsed))

##           med           iqr           max           min
## 1 9.271033 3.397883 6106.143 5.867467
```

```
quantile(duration$time_elapsed, c(.25,.75))
```

```
##           25%           75%  
## 7.959242 11.357125
```

Demographic Info

```
demo = comb_data %>% filter(typeoftrial %in% c("demo") & response != "0") %>%  
  separate(response, into = c("demo1", "demo2", "demo3", "demo4", "demo5"), sep = ":")
```

#age

```
age <- demo[!is.na(demo$age), ] %>% summarise(mean_age = mean(age))  
age
```

```
## mean_age  
## 1 36.15591
```

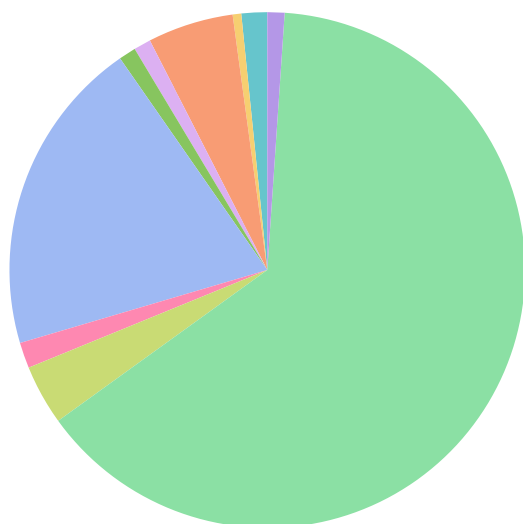
#race

```
race = demo %>%  
  filter(demo1 == "{\\\"Race\\\"}") %>%  
  select(demo2) %>%  
  mutate(demo2 = gsub("^\\[\\\"|\\\"\\]\\}$", "", demo2))
```

```
race_counts<-race %>%  
  group_by(demo2) %>%  
  mutate(demo2 = gsub("\\\"", "", demo2)) %>% rename(Race=demo2) %>%  
  count()
```

#race pie chart

```
ggplot(race_counts, aes(x = "", y = n, fill = Race)) +  
  geom_bar(stat = "identity", width = 1) +  
  coord_polar(theta = "y")+  
  theme_void()+  
  scale_fill_paletteer_d("rcartocolor::Pastel")
```

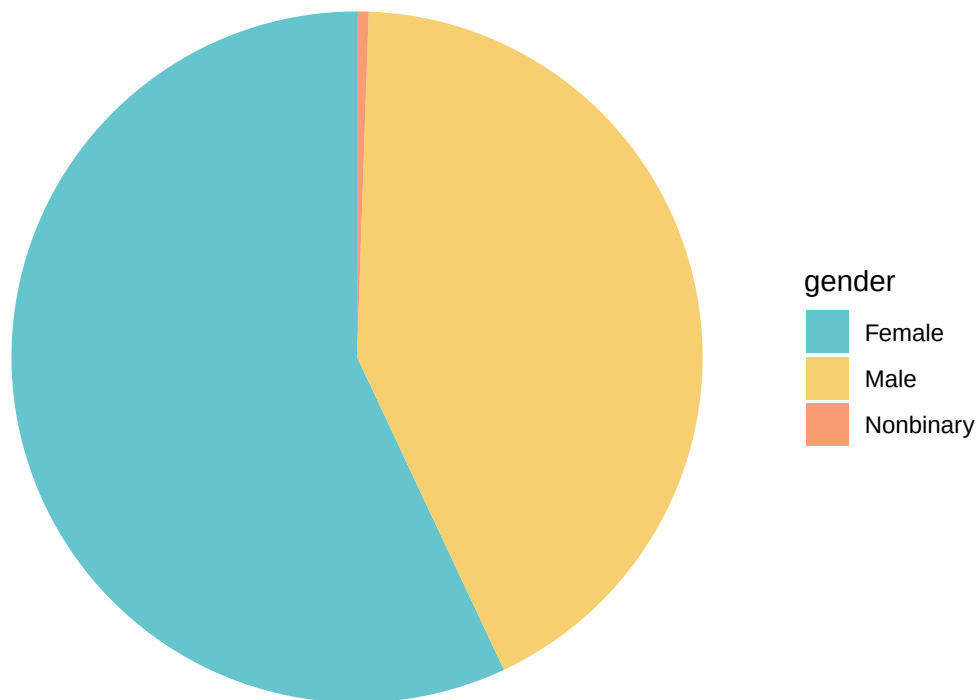


Race



```
#gender
gender_education = demo %>% select(gender, education) %>% na.omit()
gender_count <- gender_education %>%
  mutate(gender = case_when(
    gender %in% c("f", "F", "female", "Female", "woman", "women", "F", "f", "women", "Woman", "Women",
    gender %in% c("m", "M", "male", "Male", "man", "Man", "men", "Men", "MALE") ~ "Male",
    gender == "nonbinary" ~ "Nonbinary",
    TRUE ~ gender
  )) %>%
  group_by(gender) %>% count()

#gender pie chart
ggplot(gender_count, aes(x = "", y = n, fill = gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  theme_void() +
  scale_fill_paletteer_d("rcartocolor::Pastel")
```



```
#education
gender_education %>% group_by(education) %>%
  count()
```

```
## # A tibble: 13 x 2
## # Groups:   education [13]
##   education      n
##   <int> <int>
## 1         4      2
## 2         8      1
## 3        10      1
## 4        12     46
## 5        13     13
## 6        14     24
## 7        15     16
## 8        16     48
## 9        17      9
## 10       18     21
## 11       20      1
## 12       21      2
## 13       25      2
```

```
mean_ed<-gender_education[!is.na(gender_education$education), ] %>% summarise(mean_education = mean(education))
mean_ed
```

```
##   mean_education
```

1 14.69892